

Aryamaan Jain

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EDUCATION

PhD in Computer Science

Inria, Université Côte d'Azur

Oct 2023 – Sep 2026 (*Expected*)

Sophia Antipolis, France

Lab: GraphDeco · Advisor: Dr. Guillaume Cordonnier

BTech(Honours) + MS(Research) in Computer Science & Engineering

IIT Hyderabad

Jul 2019 – Jul 2023

Hyderabad, India

Lab: CVIT · Advisor: Dr. Avinash Sharma · Co-advisor: Dr. KS Rajan

CGPA: 9.55/10.0 · Specialization: Artificial Intelligence

EXPERIENCE

Visiting Researcher

Computer Science, Purdue University

Apr 2026 – May 2026

West Lafayette, USA

Collaborating with Prof. Bedrich Benes for generative 3D reconstruction of natural scenes. Visit funded by the DocWalker international mobility grant.

Research Intern

GraphDeco, Inria

Mar 2023 – Oct 2023

Sophia-Antipolis, France

Developed a physically based terrain erosion simulation accelerated with a GPU implementation and learning based super-resolution.

Research Assistant

CVIT and IHub-Data, IIT Hyderabad

Aug 2022 – Jan 2023

Hyderabad, India

Worked on 3D reconstruction of roads with LiDAR data from vehicles using the ICP algorithm.

Summer Intern

Wells Fargo (Strategy, Digital & Innovation)

May 2022 – Jul 2022

Bengaluru, India

Developed a VR Banking prototype on Oculus Quest 2. Created optimized 3D assets/scenes and integrated LLMs for user interaction via AWS.

Teaching Assistant

IIT Hyderabad

Jan 2021 – Oct 2022

Hyderabad, India

- Foundations of Modern Machine Learning (IHub-Data), Jan 2022 - Oct 2022.
- Computer Graphics, Spring 2022.
- Computer System Organisation, Spring 2021.

SELECTED PUBLICATIONS

1. Pixels2Peaks: Converting Terrain Images to Heightmaps

Aryamaan Jain, James Gain, Guillaume Cordonnier

SIGGRAPH 2026 (ACM Transactions on Graphics)

Method to infer 3D terrain heightmaps from single images. It utilizes guided diffusion models to generate occluded geometry with physical consistencies.

2. Arenite: A Physics-based Sandstone Simulator

Zhanyu Yang, Aryamaan Jain, Guillaume Cordonnier, Marie-Paule Cani, Zhaopeng Wang, Bedrich Benes

SIGGRAPH 2025 (ACM Transactions on Graphics)

GPU accelerated physics framework that generates sandstone formations by simulating the interplay of structural stress, erosion, and deposition.

3. **FastFlow: GPU Acceleration of Flow and Depression Routing for Landscape Simulation**

Aryamaan Jain, Bernhard Kerbl, James Gain, Brandon Finley, Guillaume Cordonnier

Pacific Graphics 2024 (Computer Graphics Forum)

Best Paper Award

GPU based algorithm that simulates how water flows and pools across terrain. It achieves up to 52× speedup over current methods, enabling interactive terrain simulation.

4. **Efficient Debris-flow Simulation for Steep Terrain Erosion**

Aryamaan Jain, Bedrich Benes, Guillaume Cordonnier

SIGGRAPH 2024 (ACM Transactions on Graphics)

GPU erosion algorithm incorporating a novel mathematical formulation to accurately simulate steep-slope erosion and deposition.

5. **Learning Based Infinite Terrain Generation with Level of Detailing**

Aryamaan Jain, Avinash Sharma, KS Rajan

3DV, 2024

GAN based generative framework using image completion and super-resolution to create infinite terrains with efficient quad-tree integration.

TECHNICAL SKILLS

- **Languages:** Python, C, C++, CUDA, GLSL
- **Core Libraries:** PyTorch, OpenGL, OpenCV, OpenMP, STL
- **Tools & Software:** Houdini, Blender, Unity, Terragen, Git, L^AT_EX, Bash

HONORS & AWARDS

- Recipient of the DocWalker international mobility grant (2026).
- Pierre Laffitte prize finalist (2025).
- Best paper award at Pacific Graphics (2024).
- Dean's/Merit list recipient, all semesters at IIIT Hyderabad.

ACADEMIC SERVICE

Reviewer: SIGGRAPH 2026, ICVGIP 2025.